

Dr. Akash Kumar

Remote Sensing & Crop Phenology Researcher

Earth Observation • PhenoCam • Geospatial Analytics

🏛️ Civil Engineering Dept., IIT Roorkee

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Education

Jul 2022 – Jul 2026

Ph.D. • Geospatial Engineering

🏛️ Indian Institute of Technology Roorkee

Thesis: Multi-scale analysis of crop phenology using PhenoCam & satellite data

Sep 2020 – May 2022

M.Tech • Geoinformatics

🏛️ Maulana Azad National Institute of Technology, Bhopal

Thesis: A web-GIS based approach for mapping rice residue burning and generated pollutants over Punjab • Branch Gold Medalist; 9.51 CGPA

2014 – 2018

B.Tech • Civil Engineering

🏛️ Meerut Institute of Engineering & Technology, Meerut

Project: Assessment of Structural Components and Construction Methods of Flyovers

Experience

Indian Institute of Technology Roorkee 📍 IITR, Uttarakhand, India

July 2026 – Present

Project Fellow (post-doc)

Project: "AI based Geo-enabled crop insurance farm equipment renting solutions under Bhoomi-Cam pvt. Ltd., Start-up. Sponsored by IIT Bhilai Innovation and Technology Foundation, Raipur, Chhattisgarh

Bhoomicam Pvt Ltd 📍 IIT Roorkee

Apr 2025 – Present

Co-Director

Handling operations and B2B partnerships.

Jun 2023 – Apr 2025

Chief Remote Sensing Scientist

Led the technical team in building geospatial solutions for crop monitoring.

Bharat Heavy Electricals Limited (BHEL) 📍 Yelahanka, Bengaluru

Dec 2018 – Sep 2019

Civil Site Engineer

Full-time, 10-month role at the 1 x 370 MW Yelahanka power plant; supervised civil construction, quality control, and on-site design coordination.

Publications

Peer-Reviewed Journals

- [1] **Akash Kumar** and Siddhartha Khare (2026). Evaluation of PlanetScope and Sentinel-2 Derived Wheat Crop Phenology Using Infrared-Enabled PhenoCam and Field Observations. *Ecological Informatics*, 103821. [10.1016/j.ecoinf.2026.103821](https://doi.org/10.1016/j.ecoinf.2026.103821).
- [2] **Akash Kumar**, Siddhartha Khare, and Sergio Rossi (2025). PhenoAI: A deep learning Python framework to process close-range time-lapse PhenoCam data. *Ecological Informatics*, (88), 103134. [10.1016/j.ecoinf.2025.103134](https://doi.org/10.1016/j.ecoinf.2025.103134).

- [3] **Akash Kumar**, Ganesh Sai Sivani Noolu, Siddhartha Khare, and Sergio Rossi (2025). A comparison of PhenoCam and satellite indices with in-situ observations for black spruce in the boreal forest of Quebec, Canada. *Ecological Informatics*, (88), 103137. [10.1016/j.ecoinf.2025.103137](https://doi.org/10.1016/j.ecoinf.2025.103137).

— International Conferences

- [1] **Akash Kumar** and Siddhartha Khare (2026). An Automated Workflow for Extracting Land Surface Phenology from PhenoCam Time-Lapse Images. *Asia Oceania Geosciences Society (AOGS 2026)*. Accepted and presentation scheduled in August 2026. Fukuoka, Japan, 2026.
- [2] **Akash Kumar** and Siddhartha Khare (2026). Establishing PhenoCam-based Monitoring of Wheat and Rice Phenology in India with Satellite and Meteorological Data. *EGU General Assembly 2026*. Vienna, Austria, 2026. [10.5194/egusphere-egu26-14674](https://doi.org/10.5194/egusphere-egu26-14674).
- [3] **Akash Kumar**, Noolu G. S. Sivani, Siddhartha Khare, and Sergio Rossi (2024). Linking PhenoCam Derived Phenology With Satellite and Field Observations in the Evergreen Forest. *2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*. 2024. [10.1109/InGARSS61818.2024.10984047](https://doi.org/10.1109/InGARSS61818.2024.10984047).

— Book Chapter

- [1] Tushar Bharadwaj, **Akash Kumar**, and Siddhartha Khare (2023). Smart agriculture platforms: Case studies. *Technological Prospects and Social Applications of Society 5.0*. CRC Press, 2023, 169–176. [10.1201/9781003324720-15](https://doi.org/10.1201/9781003324720-15).

— Manuscripts Under Review & Preparation

- [1] **Akash Kumar**, S. Goswami, and Siddhartha Khare (2026). *Deep Learning-Based Field Segmentation and Multitemporal Sentinel-2 Analysis for Crop Yield Estimation in Smallholder Systems of India*. In preparation. 2026.
- [2] **Akash Kumar**, Zainab Khan, Siddhartha Khare, Umesh Kumar Saxena, and Purushottam Kumar (2026). Integrating PhenoCam and AWS Data to Quantify Stage-Specific Climatic Impacts on Rice Phenology. *Remote Sensing Applications: Society and Environment*. [10.2139/ssrn.6087391](https://doi.org/10.2139/ssrn.6087391). Under review.

— Conferences & Workshops

— Conference Participation

Aug 2026	Asia Oceania Geosciences Society Annual Meeting (AOGS 2026) 📍 Fukuoka, Japan <i>Scheduled</i> Accepted: automated PhenoCam time-lapse image workflow
May 2026	EGU General Assembly 2026 (EGU26) 📍 Vienna, Austria <i>Poster presentation</i> post
Dec 2025	GeoSmart India Conference & Expo 📍 Bharat Mandapam, New Delhi <i>Presented</i> post
Feb 2025	Workshop on Synthetic Aperture Radar (SAR) 📍 IIT Tirupati <i>Attended</i>
Dec 2024	India Geoscience and Remote Sensing Symposium (InGARSS) 📍 NIT Goa <i>Paper presentation</i> post
Aug 2023	2nd Roorkee Water Conclave 📍 NIH Roorkee <i>Attended</i>
Oct 2022	Natural Hazard Symposium for the Indian Himalayas 📍 IIT Roorkee <i>Attended</i>

— Delivered & Conducted

- May 2026 **Working with Google Earth Engine** 📍 GNEC, IIT Roorkee
Invited lecture IN-SPACe Short-Term Skill Development Course [🔗 post](#)
- Jan 2023 **Modern Surveying Instrument Karyashala** 📍 IIT Roorkee
Teaching assistant
- Apr 2022 **Google Earth Engine** 📍 NIGMT New Delhi
Instructor Online
- Dec 2021 **Google Earth Engine** 📍 MANIT Bhopal
Instructor Water Resource Dept.

— Fellowships & Grants

- 2026 **AOGS reduced registration fee for the 23rd Annual Meeting**
For presentation at AOGS 2026 📍 Fukuoka, Japan
- 2026 **IIT Roorkee Department Development Fund (DDF)**
₹1,50,000 for presentation at EGU General Assembly 2026 📍 Vienna, Austria
- 2026 **NPTEL TA for 8-Week Course**
Honorarium – ₹40,000; repeat offering
- 2025 **NPTEL TA for 8-Week Course**
Honorarium – ₹1,00,000 [🔗 TA list](#)
- 2024 **IEEE InGARSS Travel Grant**
\$ 250 📍 NIT Goa
- 2022 **First Prize, National Geospatial Startup Competition by AABSyS**
₹50,000 [🔗 details](#)
- 2022 **SRF & MHRD Scholarship for PhD**
- 2020 **MHRD Scholarship for Post-Graduation**

— Teaching & Mentoring

— Teaching Assistantships

- 2023 – 2025 **Geospatial Engineering I (CEN-102)** 📍 IIT Roorkee
Teaching assistant Spring semesters; Dept. of Civil Engineering
- 2022 – 2025 **Geospatial Engineering II (CEN-201)** 📍 IIT Roorkee
Teaching assistant Autumn semesters; Dept. of Civil Engineering
- 2022 **Basics of Remote Sensing Technology (GI-103)** 📍 MANIT Bhopal
Teaching assistant Spring semester
- 2021 **Digital Processing of Remotely Sensed Data (GI-202)** 📍 MANIT Bhopal
Teaching assistant Autumn semester; online tutorials

— Mentoring & Field Training

- 2025 – 2026 **Digital Land Surveying & Mapping, NPTEL** 📍 IIT Roorkee
Course support Jan–Mar offerings; coordination of uploads, assignments, and queries
- 2023 – 2024 **External Academic Mentor of Mr. Dipanshu Priyadarshi, Gati Shakti Vishwavidyalaya** 📍 Vadodara
Academic mentor Spring semester; UG project on land-use/land-cover change detection

2022 – 2026

Hands-on sessions for railway, land-planning, and officer-training programs

📍 GNEC, Noida

Practical trainer IIT Roorkee Extension Centre

Dec 2022

Survey camp for second-year undergraduate students

Field camp lead Led field surveying and managed 120+ students

Technical Skills

Software

Programming	Python, R, JavaScript (Node.js, React.js)
Data & Statistics	Advanced MS Office, SAS JMP Pro, LaTeX, MS Visio, SigmaPlot
Geospatial	Google Earth Engine, QGIS, ArcGIS Pro, ERDAS IMAGINE, iGIS
Photogrammetry	Pix4D Mapper, FARO Scene, CloudCompare
GPS & Geotagging	Emlid Studio, Spectrum Survey Office, QBase
Spectral Analysis	SVC's Spectra Vista, ENVI
Civil Engineering	Autodesk AutoCAD 3D, Revit Architecture, STAAD.Pro

Instrument Handling

UAVs	IdeaForge Q6 (RGB and multispectral), Trinity F90+ Hybrid VTOL (LiDAR and oblique RGB), DJI Matrice 100, Agri-Spray Drone (quad- and hexacopter)
Sensors	PhenoCam camera, Automated Weather Station (AWS), Spectroradiometer SVC XHR-1024i
GPS	Trimble DA2, Emlid Reach RS2/RS3 RTK, Sokkia GRX2
TLS	FARO Terrestrial Laser Scanner
Surveying	Total Station (robotic and EDM), Analytical Theodolite, digital/automatic/IOP/dumpy levels

Professional Service

Peer Review

<i>Ecological Informatics</i>	31+ reviews
<i>International Journal of Applied Earth Observation and Geoinformation</i>	22+ reviews
<i>Information Processing in Agriculture</i>	5 reviews
<i>Smart Agricultural Technology</i>	2 reviews

Memberships & Affiliations

IEEE GRSS	Geoscience and Remote Sensing Society	No. 100490233; since Sep 2024
IIM	Indian Institute of Metals	ID 62930; since May 2025
EGU	European Geosciences Union	ID 723355; since Jan 2026
AOGS	Asia Oceania Geosciences Society	Reg. AOGS26-Reg-21198; since May 2026

Extracurricular Experience & Activities

2022 – Present	Represented Bhoomicam Pvt Ltd at startup programs, innovation showcases, and outreach events. > KrishiTech Synapse 2026 📍 IIT Delhi – <i>Third-round evaluation</i> > IN-SPACe Industry Connect 2026 – <i>Top 3 grantee ₹ 40,00,000 support</i> > NTRO AI Grand Challenge 2025 – <i>Top 6 nationwide ₹ 8,00,000 award</i> 🔗 details > Operation Dronagiri 2025 – <i>Top 25 ₹ 10,00,000 grant</i> > 100 Startups, 100 Days at IIT Ropar – <i>₹ 2,00,000 grant</i> > NASSCOM Technology Confluence 2024 🔗 post
Jan 2026	NRSC Offline Coding Challenge 2025, ISRO – <i>Second place</i> 🔗 post
Jul 2025	Geospatial Mapathon 2025, IIT Bombay FOSSEE – <i>Winner</i> 🔗 results
2024	UAV Pilot Certificate – IdeaForge – <i>Certified NP2317</i>
Oct 2024	NASA Space Apps Challenge 2024 – <i>Participant</i> 🔗 post
2022 – 2025	Inter-Bhawan Chess Tournament, IIT Roorkee – <i>Gold: 2022–2024 – Silver: 2025</i>
Mar 2016	Guinness World Record – World’s Biggest Orchestra – <i>Record holder guitarist; Art of Living Foundation, New Delhi</i> 🔗 details
Oct 2014	UP Open Chess Tournament (U-17) – <i>Ranked #3</i>

References

Prof. Siddhartha Khare

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